

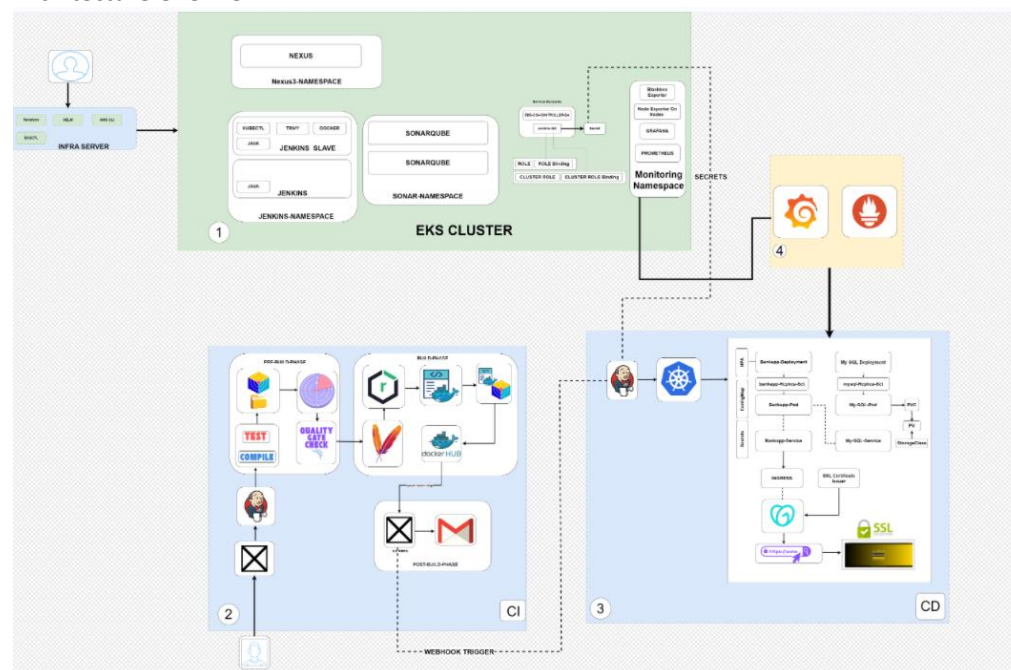
GoldenCat Bank – End-to-End DevSecOps on AWS EKS

This project implements a complete DevSecOps pipeline for a sample banking application running on AWS EKS.

It covers:

- Infrastructure & platform setup on AWS using Terraform
- CI pipeline (build, test, code quality, security scans, artifact & image publishing)
- CD pipeline to EKS with Ingress and TLS (Let's Encrypt via `cert-manager`)
- Monitoring & observability with Prometheus and Grafana

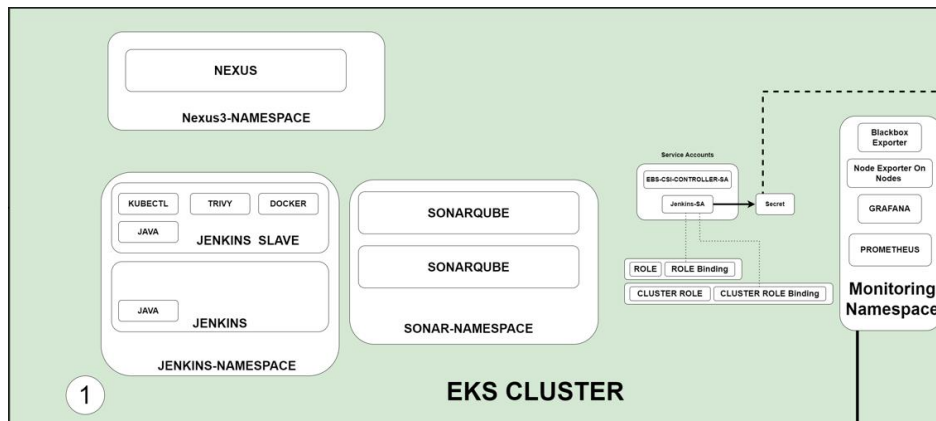
Architecture Overview



High-Level Design

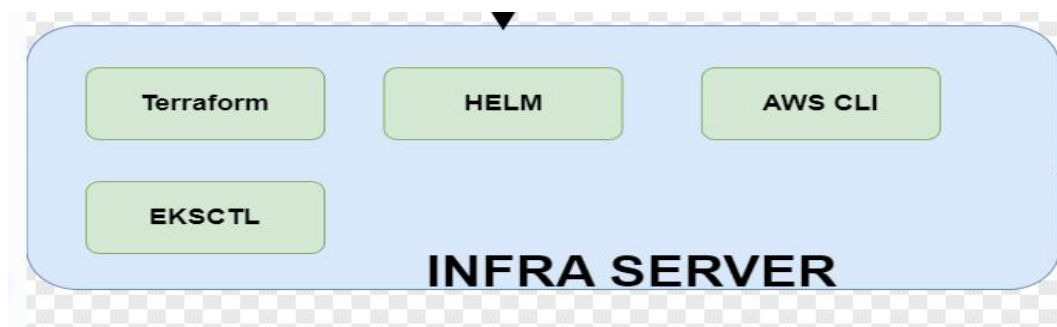
The platform runs on a single EKS cluster with multiple namespaces:

- `jenkins-namespace` – Jenkins controller & agents
- `nexus3-namespace` – Nexus3 artifact repository
- `sonar-namespace` – SonarQube for static analysis
- `webapps` – GoldenCat Bank application (bankapp + MySQL)
- `monitoring` – Prometheus & Grafana stack



An infra server (EC2) is used as the main control machine, with:

- Terraform
- kubectl
- Helm
- AWS CLI
- eksctl



The cluster runs on three worker nodes plus dedicated instances for Jenkins, SonarQube and Nexus:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
Node-1	i-0a3ccae6babe15720	Running	t2.medium	2/2 checks passed	View alarms +	us-east-1a	-	18.213.150.152
Node-2	i-019c7bc7307b15772	Running	t2.medium	2/2 checks passed	View alarms +	us-east-1a	-	3.235.16.238
Infra-server	i-0437e2d1881d8b22f	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1b	ec2-98-93-234-1.comp...	98.93.234.1
Jenkins	i-06c7536db744202b9	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1b	ec2-44-212-25-131.co...	44.212.25.131
SonarQube	i-082218219b81d04f4	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1b	ec2-44-203-90-100.co...	44.203.90.100
Nexus	i-051c754edad340f1d	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1b	ec2-34-227-161-51.co...	34.227.161.51
Node-3	i-08a32f534beef92ae	Running	t2.medium	2/2 checks passed	View alarms +	us-east-1b	-	3.92.175.221

EKS cluster overview:
powerdevops-cluster

Cluster info

Status

Active

Cluster health

0

Kubernetes version

1.34

Upgrade insights

5

Support period

Standard support until December 1, 2026

Node health issues

0

Provider

EKS

Overview

Resources

Compute

Networking

Add-ons

Access

Observability

Update history & backups

Tags

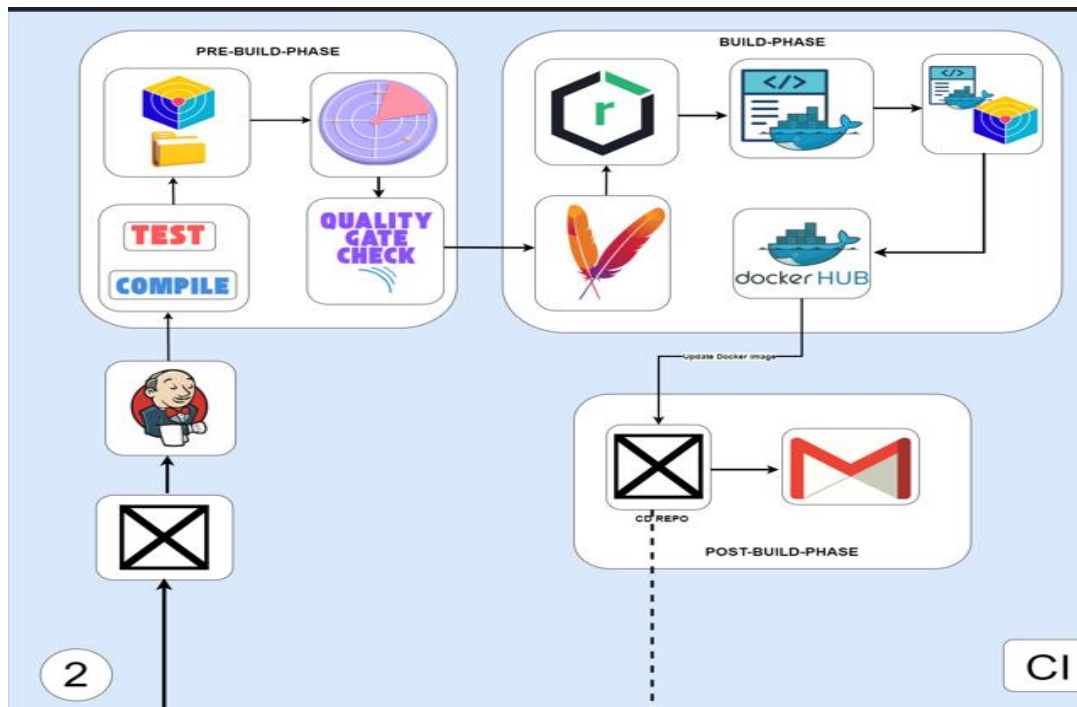
Details

API server endpoint

OpenID Connect provider URL

Created

CI Pipeline – `Mega-Project-CI`



CI Architecture

The CI pipeline is responsible for:

- Compiling and testing the Java / Spring Boot bank application
- Running static analysis and quality gates
- Scanning the filesystem and container image with Trivy
- Publishing artifacts to Nexus and container images to Docker Hub
- Triggering notifications and preparing the CD stage

Tools & Services

- Jenkins – Declarative pipeline
- GitHub – Source repository and webhooks
- Maven – Build and test
- SonarQube – Static code analysis & Quality Gate
- Trivy – Filesystem and Docker image scanning
- Nexus3 – Maven snapshot repository
- Docker Hub – Image registry
- Email – Pipeline result notification

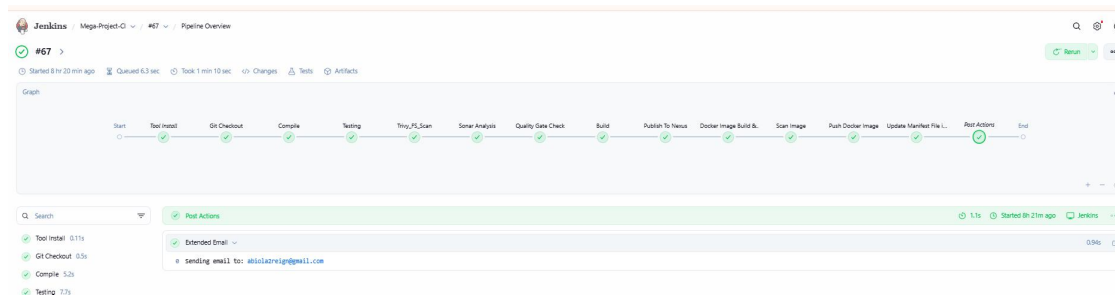
Pipeline Stages

The Jenkins CI pipeline runs the following stages:

1. Tool Install – Ensure required tools (Maven, Trivy, etc.) are available.
2. Git Checkout – Clone the `main` branch of `Mega-Project-2`.
3. Compile – Compile the code with Maven.
4. Testing – Run unit tests.
5. Trivy_FS_Scan – Scan the Jenkins workspace for vulnerabilities.

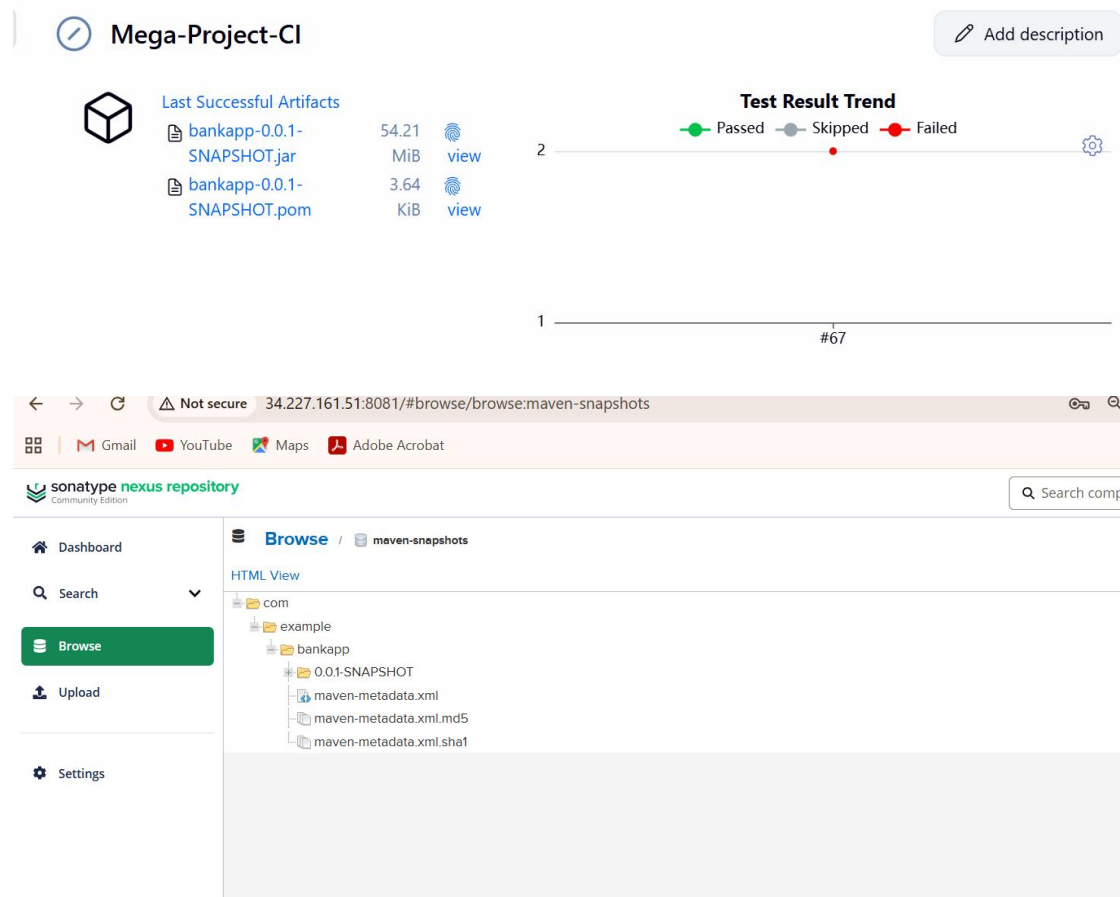
6. Sonar Analysis – Push code metrics to SonarQube project `gcbank`.
7. Quality Gate Check – Wait on SonarQube's Quality Gate result.
8. Build – Package the application JAR.
9. Publish To Nexus – Deploy artifacts to Nexus snapshot repository.
10. Docker Image Build & Scan Image – Build the Docker image and scan it with Trivy.
11. Push Docker Image – Push the image to Docker Hub.
12. Update Manifest File in Mega-Project-CD – Bump the Kubernetes image tag (e.g. `v67`).
13. Post Actions – Send email notification on success/failure.

Jenkins CI Graph



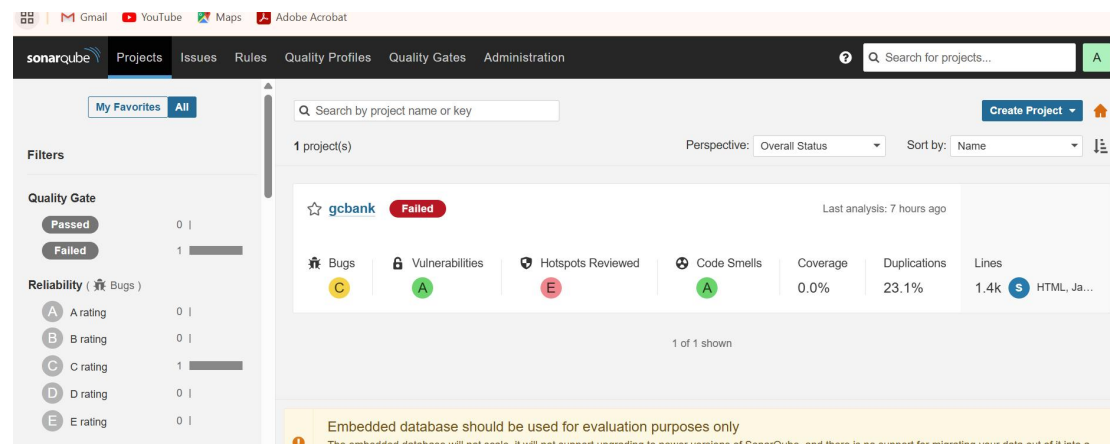
Artifacts & Nexus

Jenkins stores the compiled artifacts, while Nexus holds the Maven snapshots:



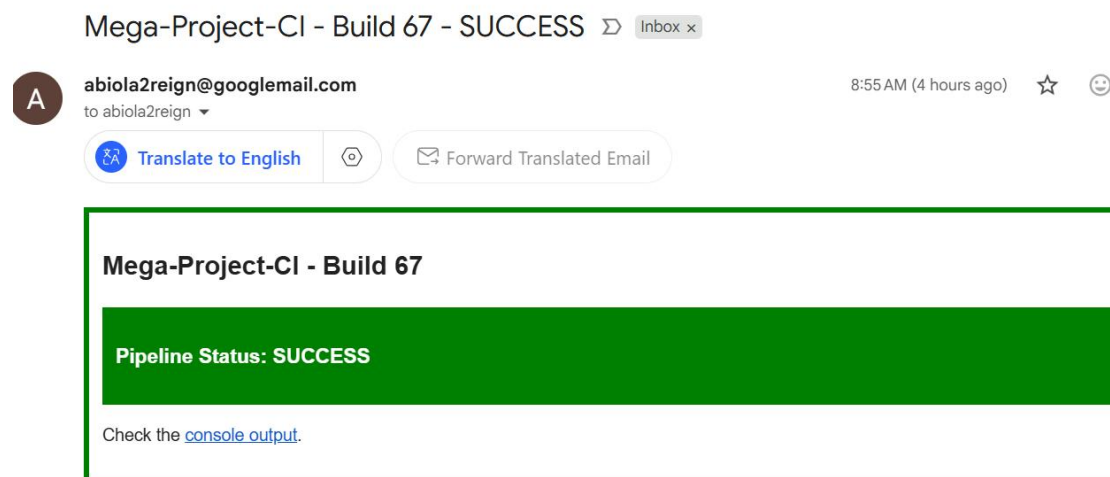
SonarQube Quality Dashboard

Static analysis and Quality Gate result for the `gcbank` project:



Email Notification

On successful pipeline completion, an email is sent summarizing the status and linking to the Jenkins console output:



CD Pipeline – `Mega-Project-CD`

3.1 CD Architecture

The CD pipeline deploys the latest Docker image to the `**webapps**` namespace in EKS and configures Ingress + TLS.

Monitoring Stack – Prometheus & Grafana

Monitoring is installed via Helm using kube-prometheus-stack:

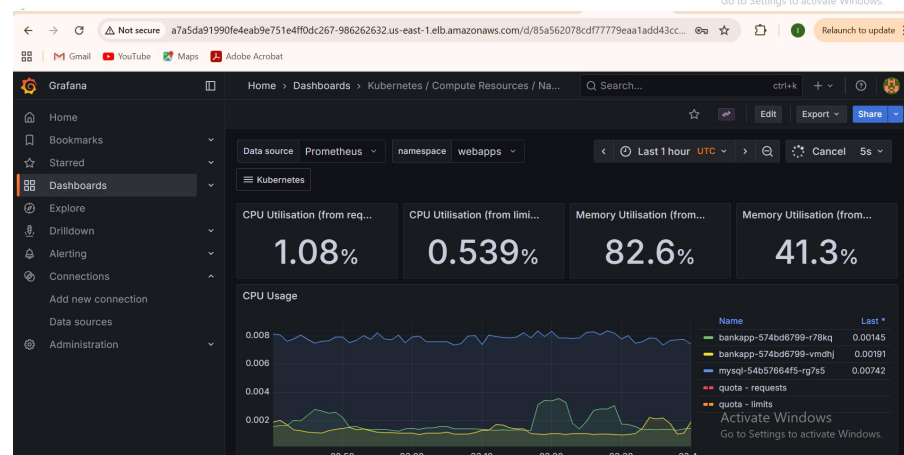
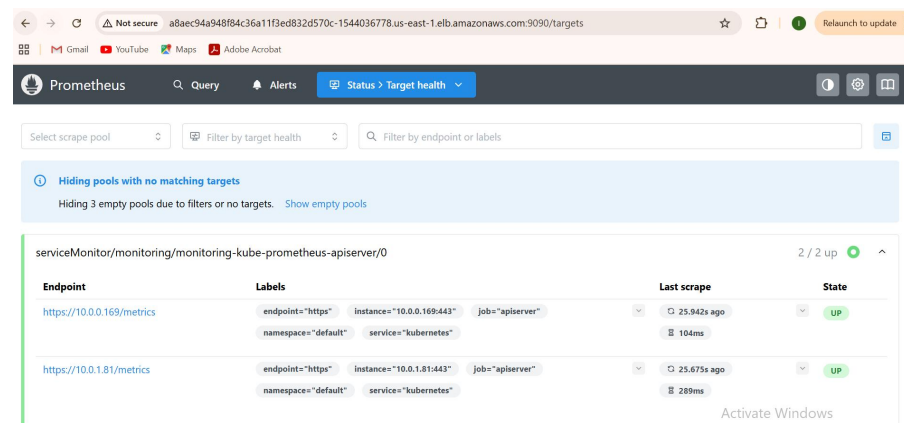
```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update
```

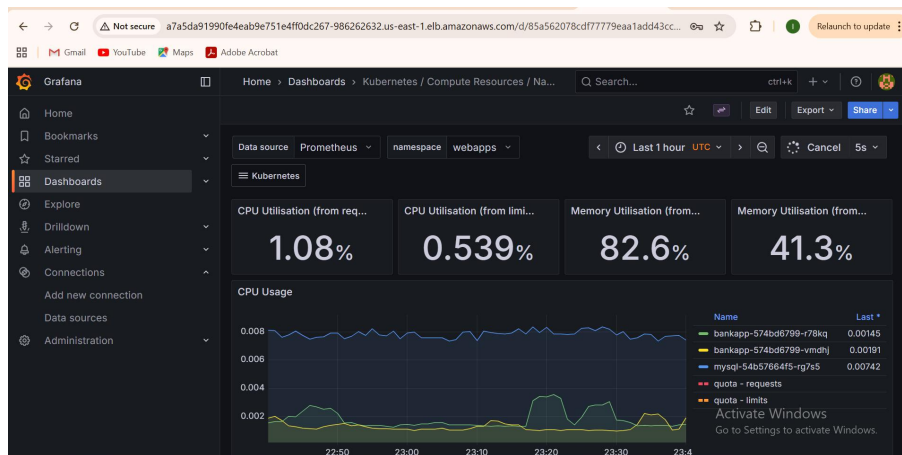
```
helm upgrade --install monitoring prometheus-community/kube-prometheus-stack \
-f values.yaml \
-n monitoring --create-namespace
```

```
ubuntu@ip-172-31-80-191:~/monitoring$ kubectl get all -n monitoring
NAME                                READY    STATUS    RESTARTS   AGE
pod/monitoring-grafana-5c7bc7b48b-vhs2q    3/3      Running   0           4m18s
pod/monitoring-kube-prometheus-operator-5cdc56d76c-q7q5b    1/1      Running   0           4m18s
pod/monitoring-kube-state-metrics-5459bf8fdf-nq8vx    1/1      Running   0           4m18s
pod/monitoring-prometheus-node-exporter-45z8j    1/1      Running   0           4m18s
pod/monitoring-prometheus-node-exporter-x9g7t    1/1      Running   0           4m18s
pod/monitoring-prometheus-node-exporter-z8crr    1/1      Running   0           4m18s
pod/prometheus-monitoring-kube-prometheus-prometheus-0    2/2      Running   0           4m13s

NAME                                PORT(S)          TYPE          AGE      CLUSTER-IP      EXTERNAL-IP
service/monitoring-grafana          80:30741/TCP      LoadBalancer  4m18s     172.20.224.133   a7a5da91990fe4eab9e751e4ff0dc267-986262632.us-east-1.elb.amazonaws.com
service/monitoring-kube-prometheus-operator    443/TCP      ClusterIP     4m18s     172.20.80.200    <none>
service/monitoring-kube-prometheus-prometheus    9090:31454/TCP,8080:30991/TCP    LoadBalancer  4m18s     172.20.16.100    a8aec94a948f84c36a11f3ed832d570c-1544036778.us-east-1.elb.amazonaws.com
service/monitoring-kube-state-metrics    8080:32298/TCP      LoadBalancer  4m18s     172.20.25.146    adc5456d0e55841b688196e174920c9a-1970335806.us-east-1.elb.amazonaws.com
service/monitoring-prometheus-node-exporter    9100:30568/TCP      LoadBalancer  4m18s     172.20.169.98    a896a396bfaab47d5a28151bde0c0748-2001084598.us-east-1.elb.amazonaws.com
service/prometheus-operated          9090/TCP      ClusterIP     4m13s     None             <none>

NAME                                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE    NODE SELECTOR
daemonset.apps/monitoring-prometheus-node-exporter    3           3           3           3           3           kubernetes.io/os=linux
```





Application Interface

The GoldenCat Bank application is exposed over HTTPS via NGINX Ingress and Let's Encrypt TLS:
<https://www.hkhpros.xyz>

The image shows a web browser window with the URL <https://www.hkhpros.xyz/login>. The page has a dark header with the text 'Goldencat Bank' and a play button icon. The main content area has a yellow background. In the center, there is a white login form with the title 'Login'. The form contains two input fields: 'Username:' and 'Password:'. Below these fields is a yellow 'Login' button. At the bottom of the form, there is a link: 'Don't have an account? [Register here](#)'.